

Reverse Engineering Assignment: Smoke and Carbon Monoxide Detector

SRIRAM ELANGO¹

¹Harvard College, Cambridge, MA 02139, USA

1. DESIGN ANALYSIS



(a) The fully assembled Kidde Smoke and Carbon Monoxide alarm, showing the external chassis, test/hush button, and battery compartment door.



(b) Internal view after removing the faceplate. The main PCB is visible alongside the battery housing and the primary sensor components.



(c) Partial disassembly of the electronics, showing the separation of the circuit board from the rear housing and the wired connections.



(d) Detailed view of the opened smoke detector sensor and the logic board responsible for gas detection.

Figure 1. Sequential teardown of a Kidde combination alarm.

Within Figure 1a, is the untouched, fully assembled Kidde Smoke and Carbon Monoxide Alarm, with a top-down external view. The ABS injection molded plastic exterior shell is clearly visible. When observing the exterior from a side view, it is possible to observe an upper and lower shell, that is attached by a strong clip-on mechanism. In order to detach the plastic shells, a flat head screwdriver was used as a lever to snap the clip-on mechanisms. Once separated, the internal electronics could be viewed, as shown in Figure 1b. The PCB board is clearly visible, with a battery attached, as well as a smoke detector module, all wired to the upper shell where there includes some more sensors and auditory alarms. Snipping the wired connections to the plastic shell, a decomposed view can be seen, with the physical exterior shell and the electronic innards separated in Figure 1c. Zooming into the PCB board, it is possible to decompose the smoke detector module further and observe even further electronic components such as LEDs, and an increased density of wired-in modules within the PCB board, seen in Figure 1d. Overall, this design is quite efficient for mass production. The exterior injection-molded ABS plastic is as optimal as it gets for mass manufacturing for strong, durable quality parts. By using a clip on mechanism to attached the upper and lower shell, it reduces assembly time and unnecessary screws, rather relying on the internal strength of the already cheap plastic shells. The PCB board (also easy to mass produce) is attached by light sautering and a small number of screws, thus minimizing cost and potential points of failure. The more expensive components are likely the smoke sensor itself, but that can be purchased from an external vendor, and the PCB board is already designed to house the detector, making it easy to assemble the PCB board together. Overall, the Smoke Alarm is very efficiently designed for mass manufacturing, with easy to assemble components already fitted to each other in the design phase, with minimal points of failure.